

Autonomous Folkrace Car – Project Overview

Project Aim

To design and develop a fully autonomous 4-wheel racing vehicle capable of navigating a circular track, avoiding obstacles, handling curves, and completing laps with minimal external control. The vehicle is intended for use in academic competitions and showcases key embedded systems and robotics principles.

Key Features

- **Autonomous Navigation:** Follows a defined track and adapts to obstacles in real time.
 - **Dynamic Speed Control:** Adjusts motor speed based on obstacle proximity for safe and efficient travel.
 - **Curve Handling:** Uses side ultrasonic sensors to steer through curves dynamically.
 - **Bluetooth Start/Stop:** Controlled remotely to begin or halt operation via Bluetooth communication.
 - **Real-Time Sensing:** Front and side obstacle detection using ultrasonic sensors for better decision-making.
 - **Lightweight Design:** Optimized for speed, maneuverability, and power efficiency.
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Main Components

Component	Function
ESP32 Dev Module	Central microcontroller; handles logic, control, and Bluetooth
Ultrasonic Sensors (5x)	Obstacle detection (Front, Left, Right, Mid-Left, Mid-Right)
Motor Driver (L298N)	Controls the motors based on ESP32 PWM outputs
DC Gear Motors (x2)	Drives 4 wheels via gear mechanism (dual motor drive)
Bluetooth Module	Built-in ESP32 module used to send start/stop commands
Power Supply	7.4V Li-Ion battery or 2x 18650 cells with voltage regulation
Chassis	Lightweight frame designed to support components and movement

System Development Phases

1. Planning & Research

- Analyzed rules and constraints of the folkrace competition
- Studied required speed, turning radius, and track design
- Defined hardware requirements for sensor placement and control logic

2. Hardware Design

- Selected suitable motors, sensors, and microcontroller
- Designed power circuit and wiring layout
- Built and tested custom chassis for balance and traction

3. Firmware Development

- Implemented ultrasonic reading and filtering
- Developed obstacle avoidance logic based on distance thresholds
- Added speed control based on proximity to obstacles
- Implemented curve handling using side sensor comparisons
- Added Bluetooth control logic (start/stop)

4. Testing & Tuning

- Calibrated sensor ranges and delay timing
- Tuned motor speed (PWM values) for different situations
- Field-tested on multiple track configurations

5. Final Integration

- Verified autonomous lap completion
- Ensured smooth handling of curves and stops
- Validated system behavior under various obstacle scenarios